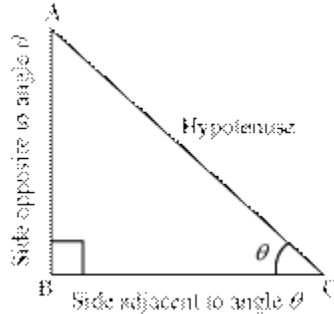


## Introduction to Trigonometry

- Trigonometric Ratio



$$\sin \theta = \frac{\text{Opposite side}}{\text{Hypotenuse}} = \frac{AB}{AC}$$

$$\cos \theta = \frac{\text{Adjacent side}}{\text{Hypotenuse}} = \frac{BC}{AC}$$

$$\tan \theta = \frac{\text{Opposite side}}{\text{Adjacent side}} = \frac{AB}{BC}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{\text{Hypotenuse}}{\text{Opposite side}} = \frac{AC}{AB}$$

$$\sec \theta = \frac{1}{\cos \theta} = \frac{\text{Hypotenuse}}{\text{Adjacent side}} = \frac{AC}{BC}$$

$$\cot \theta = \frac{1}{\tan \theta} = \frac{\text{Adjacent side}}{\text{Opposite side}} = \frac{BC}{AB}$$

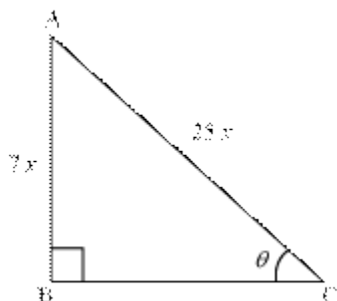
$$\text{Also, } \tan \theta = \frac{\sin \theta}{\cos \theta}, \cot \theta = \frac{\cos \theta}{\sin \theta}$$

If one of the trigonometric ratios of an acute angle is known, then the remaining trigonometric ratios of the angle can be calculated.

**Example:**

If  $\sin \theta = \frac{7}{25}$ , then find the value of  $\sec \theta(1 + \tan \theta)$ .

**Solution:**



It is given that  $\sin \theta = \frac{7}{25}$

$$\sin \theta = \frac{AB}{AC} = \frac{7}{25}$$

$\Rightarrow AB = 7x$  and  $AC = 25x$ , where  $x$  is some positive integer

By applying Pythagoras theorem in  $\triangle ABC$ , we get:

$$AB^2 + BC^2 = AC^2$$

$$\Rightarrow (7x)^2 + BC^2 = (25x)^2$$

$$\Rightarrow 49x^2 + BC^2 = 625x^2$$

$$\Rightarrow BC^2 = 625x^2 - 49x^2$$

$$\Rightarrow BC = \sqrt{576}x = 24x$$

$$\therefore \sec \theta = \frac{\text{Hypotenuse}}{\text{Adjacent side}} = \frac{25}{24}$$

$$\tan \theta = \frac{\text{Opposite side}}{\text{Adjacent side}} = \frac{7}{24}$$

$$\therefore \sec \theta (1 + \tan \theta) = \frac{25}{24} \left( 1 + \frac{7}{24} \right) = \frac{25}{24} \times \frac{31}{24} = \frac{775}{576}$$

- Use trigonometric ratio in solving problem.

Example:

If  $\tan \theta = \frac{3}{5}$ , then find the value of  $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$

Solution:

$$\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$$

Take  $\cos \theta$  common from numerator and denominator both

$$= \frac{\frac{\sin \theta}{\cos \theta} + 1}{\frac{\sin \theta}{\cos \theta} - 1}$$

$$= \frac{\tan \theta + 1}{\tan \theta - 1}$$

$$= \frac{\frac{3}{5} + 1}{\frac{3}{5} - 1}$$

$$= \frac{\frac{3+5}{5}}{\frac{3-5}{5}}$$

$$= \frac{8}{-2}$$

$$= -4$$

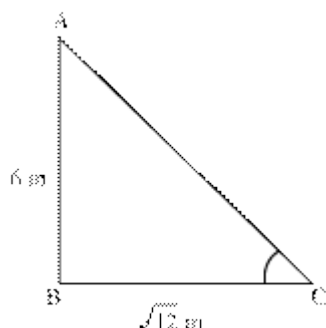
• **Trigonometric Ratios of some specific angles**

| $q$                      | 0           | 30°                  | 45°                  | 60°                  | 90°         |
|--------------------------|-------------|----------------------|----------------------|----------------------|-------------|
| $\sin q$                 | 0           | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1           |
| $\cos q$                 | 1           | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        | 0           |
| $\tan q$                 | 0           | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | Not defined |
| $\operatorname{cosec} q$ | Not defined | 2                    | $\sqrt{2}$           | $\frac{2}{\sqrt{3}}$ | 1           |
| $\sec q$                 | 1           | $\frac{2}{\sqrt{3}}$ | $\sqrt{2}$           | 2                    | Not defined |
| $\cot q$                 | Not defined | $\sqrt{3}$           | 1                    | $\frac{1}{\sqrt{3}}$ | 0           |

**Example 1:**

$\triangle ABC$  is right-angled at B and  $AB = 6$  m,  $BC = \sqrt{12}$  m. Find the measure of  $\angle A$  and  $\angle C$ .

**Solution:**



$$AB = 6 \text{ m,}$$

$$BC = \sqrt{12} \text{ m} = 2\sqrt{3} \text{ m}$$

$$\tan C = \frac{\text{Opposite side}}{\text{Adjacent side}} = \frac{AB}{BC} = \frac{6}{2\sqrt{3}} = \sqrt{3}$$

$$\Rightarrow \tan C = \tan 60^\circ \quad \left[ \because \tan 60^\circ = \sqrt{3} \right]$$

$$\Rightarrow \angle C = 60^\circ$$

$$\therefore \angle A = 180^\circ - (90 + 60) = 30^\circ$$

**Example 2:**

Evaluate the expression

$$4(\cos^3 60^\circ - \sin^3 30^\circ) + 3(\sin 30^\circ - \cos 60^\circ)$$

**Solution:**

$$4(\cos^3 60^\circ - \sin^3 30^\circ) + 3(\sin 30^\circ - \cos 60^\circ)$$

$$= 4 \left[ \left( \frac{1}{2} \right)^3 - \left( \frac{1}{2} \right)^3 \right] + 3 \left( \frac{1}{2} - \frac{1}{2} \right)$$

$$= 4 \times 0 + 3 \times 0 = 0 + 0 = 0$$

## • Trigonometric Ratios of Complementary Angles

$$\sin(90^\circ - \theta) = \cos \theta$$

$$\cos(90^\circ - \theta) = \sin \theta$$

$$\tan(90^\circ - \theta) = \cot \theta$$

$$\cot(90^\circ - \theta) = \tan \theta$$

$$\operatorname{cosec}(90^\circ - \theta) = \sec \theta$$

$$\sec(90^\circ - \theta) = \operatorname{cosec} \theta$$

Where  $\theta$  is an acute angle.

**Example 1:** Evaluate the expression

$$\sin 28^\circ \sin 30^\circ \sin 54^\circ \sec 36^\circ \sec 62^\circ$$

**Solution:**

$$\begin{aligned}
& \sin 28^\circ \sin 30^\circ \sin 54^\circ \sec 36^\circ \sec 62^\circ \\
&= (\sin 28^\circ \sec 62^\circ)(\sin 54^\circ \sec 36^\circ) \sin 30^\circ \\
&= \{\sin 28^\circ \operatorname{cosec}(90^\circ - 62^\circ)\} \{\sin 54^\circ \operatorname{cosec}(90^\circ - 36^\circ)\} \sin 30^\circ \\
&= (\sin 28^\circ \operatorname{cosec} 28^\circ)(\sin 54^\circ \operatorname{cosec} 54^\circ) \sin 30^\circ \\
&= \left( \sin 28^\circ \frac{1}{\sin 28^\circ} \right) \left( \sin 54^\circ \frac{1}{\sin 54^\circ} \right) \times \frac{1}{2} \\
&= \frac{1}{2}
\end{aligned}$$

**Example 2:** Evaluate the expression

$$4\sqrt{3}(\sin 40^\circ \sec 30^\circ \sec 50^\circ) + \frac{\sin^2 34^\circ + \sin^2 56^\circ}{\sec^2 31^\circ - \cot^2 59^\circ}$$

**Solution:**

$$\begin{aligned}
& 4\sqrt{3}(\sin 40^\circ \sec 30^\circ \sec 50^\circ) + \frac{\sin^2 34^\circ + \sin^2 56^\circ}{\sec^2 31^\circ - \cot^2 59^\circ} \\
&= 4\sqrt{3}[\sec 30^\circ(\sin 40^\circ \sec 50^\circ)] + \frac{\sin^2 34^\circ + \sin^2(90^\circ - 56^\circ)}{\sec^2 31^\circ - \tan^2(90^\circ - 59^\circ)} \\
&\quad [\because \cos(90^\circ - \theta) = \sin \theta, \tan(90^\circ - \theta) = \cot \theta] \\
&= 4\sqrt{3}[\sec 30^\circ \sin 40^\circ \operatorname{cosec}(90^\circ - 50^\circ)] + \frac{\sin^2 34^\circ + \cos^2 34^\circ}{\sec^2 31^\circ - \tan^2 31^\circ} \\
&= 4\sqrt{3}\left[\frac{2}{\sqrt{3}} \sin 40^\circ \operatorname{cosec} 40^\circ\right] + \frac{1}{1} \\
&= 8 + 1 = 9
\end{aligned}$$

### • Trigonometric Identities

1.  $\cos^2 A + \sin^2 A = 1$
2.  $1 + \tan^2 A = \sec^2 A$
3.  $1 + \cot^2 A = \operatorname{cosec}^2 A$

**Example:**

If  $\cos \theta = \frac{5}{7}$ , find the value of  $\cot \theta + \operatorname{cosec} \theta$

**Solution:**

We have,  $\cos \theta = \frac{5}{7}$

Now,  $\sin^2 \theta + \cos^2 \theta = 1$

$$\therefore \sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$= \sqrt{1 - \left(\frac{5}{7}\right)^2}$$

$$= \sqrt{\frac{49-25}{49}} = \frac{2\sqrt{6}}{7}$$

$$\therefore \operatorname{cosec} \theta = \frac{7}{2\sqrt{6}}$$

$$\text{Also, } \cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\frac{5}{7}}{\frac{2\sqrt{6}}{7}} = \frac{5}{2\sqrt{6}}$$

$$\therefore \cot \theta + \operatorname{cosec} \theta = \frac{5}{2\sqrt{6}} + \frac{7}{2\sqrt{6}}$$

$$= \frac{12}{2\sqrt{6}} = \frac{6}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}}$$

$$= \sqrt{6}$$

- **Use of trigonometric identities in proving relationships involving trigonometric ratio.**

**Example:** Prove the following identities

$$\tan^2 \theta + \cot^2 \theta + 2 = \sec^2 \theta \operatorname{cosec}^2 \theta$$

**Solution:**

We have

$$\text{LHS} = \tan^2 \theta + \cot^2 \theta + 2$$

$$= \tan^2 \theta + \cot^2 \theta + 2 \cdot \tan \theta \cdot \cot \theta \quad [\because \tan \theta \cdot \cot \theta = 1]$$

$$= (\tan \theta + \cot \theta)^2 \quad \left[ \because a^2 + b^2 + 2ab = (a + b)^2 \right]$$

$$= \left( \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right)^2$$

$$= \left( \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cdot \cos \theta} \right)^2$$

$$= \left( \frac{1}{\sin \theta \cdot \cos \theta} \right)^2$$

$$= (\sec \theta \cdot \operatorname{cosec} \theta)^2$$

$$= \sec^2 \theta \cdot \operatorname{cosec}^2 \theta$$

$$= \text{RHS}$$